Higher Reciprocity Laws

Project Report submitted

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Certificate

This is to certify that the project work entitled "*Higher Reciprocity Laws*" submitted by **Ms. Disha Atukuri** is a bonafide project work done under my supervision. It is being submitted in partial fulfillment of the requirements of Integrated Master of Science in Mathematical Sciences (I.M.Sc) degree of University of Hyderabad.

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Declaration

I, **Disha Atukuri**, hereby declare that the project work entitled "*Higher Reciprocity Laws*" is an original record of studied and bonafide work carried out by me under the guidance of Dr. Mohan Chintamani, School of Mathematics and Statistics, University of Hyderabad, Telangana, India and has not been submitted by me elsewhere for the award of any degree, diploma, title, or recognition before.

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Abstract

We study the generalization of the Quadratic Residue Character to Cubic and Biquadratic Residue Characters. We use them to define Higher Reciprocity Laws. We study the Generalized Lucas-Lehmer Test as an application of the Cubic Reciprocity Law. In doing so, we follow the paper by P. Berrizbeitia and T. G. Berry [Proc. AMS 127(7) 1923-1924]. Throughout this report, we use the notations and terminologies as in "*A Classical Introduction to Modern Number Theory*" by Kenneth Ireland and Michael Rosen.

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Chapter 1

Introduction

The Law of Reciprocity is a fundamental result in Number Theory. It was first formulated by Euler and Legendre, proved by Gauss, and partly by Legendre. Gauss was extremely proud of this result. He called it the *Theorema Aureum*, the golden theorem.

Having more than 100 proofs, the Law of Quadratic Reciprocity can qualify as one of the more complex and mind-boggling laws in the natural sciences. During the period 17th - 20th centuries, several mathematicians came up with at least two proofs for this law.

The Law of Quadratic Reciprocity provides a simple and efficient way to compute the Legendre symbol for large values. It also provides a way to determine if a congruence

$$x^2 \equiv a \pmod{p},$$

is solvable for a given prime p. However, the problem is much more difficult if the question is turned around. Suppose that a is an integer. For which primes p is the congruence

$$x^2 \equiv a \pmod{p},$$

solvable? The Law of Quadratic Reciprocity provides the answer.

One of the proofs of the Quadratic Reciprocity Law uses the notion of a quadratic Gauss Sum. The quadratic Gauss Sum can be generalized and used along with a Jacobi Sum to generalize the Quadratic Reciprocity Law to Higher Reciprocity Laws. These concepts are discussed in the following chapter.

Formulating Higher Reciprocity Laws, such as the Cubic Reciprocity Law and the Biquadratic Reciprocity Law, allows us to determine if the congruence

$$x^3 \equiv a \pmod{p}$$

and

$$x^4 \equiv a \pmod{p}$$

are solvable, which will be discussed in the subsequent chapters.

Prime numbers are an important concept in mathematics and play a significant role in various algorithms and applications. A primality test is an algorithm for determining whether an input number is prime. Unlike integer factorization, primality tests do not generally give prime factors, only stating whether the input number is prime. As an application of the Cubic Reciprocity Law, Pedro Berrizbeitia and T. G. Berry proposed [4] a primality test that is analogous to the Lucas-Lehmer test for integers of the form $M \cdot 3^a \pm 1$.

In the subsequent chapters, we will discuss the results of the Lucas Lehmer test and use the Cubic Reciprocity Law to derive its generalized test.

Chapter 2

Gauss and Jacobi Sums

Quadratic Gauss Sums are used to prove the Law of Quadratic Reciprocity. Similarly, Generalised Gauss Sums are used to study Cubic and Biquadratic Reciprocity Laws. In this chapter, we will recall some important results from Generalised Gauss Sums and Jacobi sums that will help us generalize the Quadratic Reciprocity Law to Higher Reciprocity Laws. For detailed proofs, one can refer to [1].

First, we generalize the Quadratic Character (Legendre Symbol).

Definition 2.1 (Multiplicative Character)**.** A ***multiplicative character*** on $\mathbb{F}_p$ is a map f from $\mathbb{F}_p^*$ to the group of non-zero complex numbers that satisfy $f(ab) = f(a)f(b)\ \forall a, b \in \mathbb{F}_p^*$.

$$f : \mathbb{F}_p^* \to \mathbb{C}^*$$

The Legendre symbol is a multiplicative character of order 2.

Lemma 2.1. Let f be a multiplicative character and $a \in \mathbb{F}_p^*$. Then

1. $f(1) = 1$,
2. $f(a)$ is a $(p-1)th$ root of unity,
3. $f(a^{-1}) = f(a)^{-1} = \overline{f(a)}$.
4. If $f \neq \epsilon$, then $\sum f(t) = 0$, where the sum is over all $t \in \mathbb{F}_p$. If $f = \epsilon$, then $\sum_{t \in \mathbb{F}_p} f(t) = p$.
5. If $a \in \mathbb{F}_p^*$ and $a \neq 1$, then $\sum_f f(a) = 0$, where the summation is over all characters.
6. If $a \in \mathbb{F}_p^*$, $n | p-1$, and $x^n = a$ is not solvable, then there is a character f such that

(a) $f^n = \epsilon$,

(b) $f(a) \neq 1$.

Multiplicative characters can be used to find the number of solutions for a given equation of the form $x^n = a$.

Lemma 2.2. $N(x^n = a) = \sum_{f^n=\epsilon} f(a)$, where sum is over all the characters of order dividing n.

Next, we define generalized Gauss sums using Multiplicative Characters.

Definition 2.2 (Gauss sum)**.** Let f be a character on $\mathbb{F}_p$ and $a \in \mathbb{F}_p$. Set $g_a(f) = \sum_t f(t)\zeta^{at}$, where the sum is over all t in $\mathbb{F}_p$, and $\zeta = e^{\frac{2\pi i}{p}}$. $g_a(f)$ is called a ***Gauss sum*** on $\mathbb{F}_p$ belonging to the character f.

Lemma 2.3. The following are the properties of Gauss sums:

1. If $a \neq 0$ and $f \neq \epsilon$, we, have $g_a(f) = f(a^{-1})g_1(f)$.
2. If $a \neq 0$ and $f = \epsilon$, we have $g_a(\epsilon) = 0$.
3. If $a = 0$ and $f \neq \epsilon$, we have $g_0(f) = 0$.
4. If $a = 0$ and $f = \epsilon$, we have $g_0(\epsilon) = p$.
5. If $f \neq \epsilon$, then $|g(f)| = \sqrt{p}$.

Next, we define Jacobi Sums. Multiplicative Characters and Jacobi Sum are used to find solutions of equations of the form $x^n + y^n = 1$.

Definition 2.3 (Jacobi sum)**.** Let f and h be two characters of $\mathbb{F}_p$ and set $J(f,h) = \sum_{a+b=1} f(a)h(b)$. $J(f,h)$ is called a ***Jacobi sum***.

The following are the properties of Jacobi sums:

Lemma 2.4. Let f and h be non-trivial characters. Then,

1. $J(\epsilon,\epsilon) = p$, and $J(\epsilon,f) = 0$,
2. $J(f,f^{-1}) = -f(-1)$,
3. If $fh \neq \epsilon$, then $J(f,h) = \frac{g(f)g(h)}{g(fh)}$, and $|J(f,h)| = \sqrt{p}$.

Lemma 2.5. Suppose that $p \equiv 1 \pmod{n}$ and that f is a character of order n. Then

$$g(f)^n = f(-1)pJ(f,f)J(f,f^2)\dots J(f,f^{n-2}).$$

Corollary. *If f is a cubic character, then*

$$g(f)^3 = pJ(f,f).$$

Lemma 2.6. Suppose that $p \equiv 1 \pmod 3$, and that f is a cubic character. Set $J(f,f) = a + b\omega$, as obtained in the discussion of $N(x^3 + y^3 = 1)$, where $a, b \in \mathbb{Z}$ and $\omega = e^{\frac{2\pi i}{3}} = \frac{-1+\sqrt{-3}}{2}$. Then

1. $b \equiv 0 \pmod 3$, and
2. $a \equiv -1 \pmod 3$.

In the following chapters, we will define and prove the Cubic Reciprocity Law using the notion of Generalised Gauss Sums and Jacobi Sums.

Chapter 3

The Ring $\mathbb{Z}[\omega]$

Let $\omega = \frac{-1+\sqrt{-3}}{2}$. The elements of the ring $\mathbb{Z}[\omega]$ are of the form $a + b\omega$, a, $b \in \mathbb{Z}$. If $\alpha = a + b\omega \in \mathbb{Z}[\omega]$, then the norm of α is defined by $N(\alpha) = \alpha\overline{\alpha} = a^2 - ab + b^2$. Using the norm function, we can prove that $\mathbb{Z}[\omega]$ is an Euclidean Domain. One may refer to [1] for the proof. Here, $\overline{\alpha}$ means the complex conjugate of α.

3.1 Units and Primes in $\mathbb{Z}[\omega]$

The following lemma characterizes the units of this ring.

Lemma 3.1. $\alpha \in \mathbb{Z}[\omega]$ is a unit if and only if $N(\alpha) = 1$. The units in $\mathbb{Z}[\omega]$ are 1, -1, ω, $-\omega$, ω^2, and $-\omega^2$.

Proof. ($\Leftarrow$): If $N(\alpha) = 1 \implies \alpha\overline{\alpha} = 1 \implies \alpha$ is a unit, since $\overline{\alpha} \in \mathbb{Z}[\omega]$.
($\Rightarrow$): If α is a unit, $\exists \beta \in \mathbb{Z}[\omega]$ such that $\alpha\beta = 1$. Thus, $N(\alpha)N(\beta) = N(1) = 1$. Since $N(\alpha)$ and $N(\beta)$ are positive integers, this implies that $N(\alpha) = 1$.
Suppose that $\alpha = a + b\omega$ is a unit in $\mathbb{Z}[\omega]$. Then,

$$\begin{aligned}\alpha\overline{\alpha} &= (a + b\omega)(a + b\omega^2) \\ &= a^2 + ab\omega^2 + ab\omega + b^2\omega^3 \\ &= a^2 + ab(\omega + \omega^2) + b^2 \\ &= a^2 - ab + b^2.\end{aligned}$$

Thus,

$$\begin{aligned}1 &= a^2 - ab + b^2 \\ \implies 4 &= 4a^2 - 4ab + 4b^2 \\ \implies 4 &= (2a - b)^2 + 3b^2.\end{aligned}$$

There are two possibilities in this case:

1. $(2a - b) = \pm 1$ and $b = \pm 1$.

2. $(2a - b) = \pm 2$ and $b = 0$.
   Upon considering both cases, we get the following units in $\mathbb{Z}[\omega]$:

   (a) $2a - b = 1,\ b = 1 \implies a = 1, b = 1 \implies \alpha = 1 + \omega = -\omega^2$.

   (b) $2a - b = -1,\ b = -1 \implies a = -1, b = -1 \implies \alpha = \omega^2$.

   (c) $2a - b = 1,\ b = -1 \implies a = 0, b = -1 \implies \alpha = -\omega$.

   (d) $2a - b = -1,\ b = 1 \implies a = 0, b = 1 \implies \alpha = \omega$.

   (e) $2a - b = -2,\ b = 0 \implies a = -1, b = 0 \implies \alpha = -1$.

   (f) $2a - b = 2,\ b = 0 \implies a = 1, b = 0 \implies \alpha = 1$.

Thus, the units in $\mathbb{Z}[\omega]$ are 1, -1, ω, $-\omega$, ω^2, and $-\omega^2$. □

Primes in $\mathbb{Z}$ may not be primes in $\mathbb{Z}[\omega]$. From now on, we will refer to primes in $\mathbb{Z}$ as rational primes and primes in $\mathbb{Z}[\omega]$ simply as primes. The following three lemmas help us complete the primes classification in $\mathbb{Z}[\omega]$.

Lemma 3.2. If π is a prime in $\mathbb{Z}[\omega]$, then there is a rational prime p such that $N(\pi) = p$ or p^2. In the former case, π is not associate to a rational prime; in the latter case, π is associate to p.

Proof. We have that $N(\pi) = n > 1$, since π is a prime and not a unit. This implies that

$$N(\pi) = \pi\overline{\pi} = n = p_1 p_2 \dots p_n,$$

where each p_i is a rational prime. Thus $\exists p_i$ such that $\pi | p_i$. Let that p_i be p. This implies $\pi | p$ for some rational prime p. If $p = \pi r$, $r \in \mathbb{Z}[\omega]$, then,

$$\begin{aligned} N(p) &= N(\pi)N(r) \\ \implies p^2 &= N(\pi)N(r). \end{aligned}$$

This gives us three possible cases:

1. $N(\pi) = p^2,\ N(r) = 1$.

2. $N(\pi) = p,\ N(r) = p$.

3. $N(\pi) = 1,\ N(r) = p^2$.

The third case gets rejected because $N(\pi) = 1 \implies \pi$ is a unit, which contradicts our assumption that π is not a unit.
In the first case, if $N(\pi) = p^2$, then r is a unit and we have $p = \pi r$; if r is a unit, then we say $p \sim \pi$.
In the second case, if $N(\pi) = p$, then r is not a unit since $N(r) = p$. Thus, p is not associate to π. If we consider some rational prime q such that $q = \pi b$, and b is a unit, then,

$$N(q) = N(\pi)N(b) \implies q^2 = p.1.$$

This is impossible since a rational prime cannot be a square of another rational prime. □

Lemma 3.3. If $\pi \in \mathbb{Z}[\omega]$ is such that $N(\pi) = p$, a rational prime, then π is a prime in $\mathbb{Z}[\omega]$.

Proof. We will prove this by contradiction. Assume that π is not a prime in $\mathbb{Z}[\omega]$. Then, $\pi = ab \implies p = N(a)N(b)$, which is not possible since p is prime in $\mathbb{Z}[\omega]$, so p cannot be written as a product of two integers. Thus, π is a prime in $\mathbb{Z}[\omega]$. □

Lemma 3.4. Suppose that p and q are rational primes. If $q \equiv 2 \pmod 3$, then q is a prime in $\mathbb{Z}[\omega]$. If $p \equiv 1 \pmod 3$, then $p = \pi\overline{\pi}$, where π is a prime in $\mathbb{Z}[\omega]$. Finally, $3 = -\omega^2(1-\omega)^2$, and $1-\omega$ is a prime in $\mathbb{Z}[\omega]$.

Proof. We consider the three cases separately.

Case 1. Suppose that q was not a prime in $\mathbb{Z}[\omega]$. Then, $q = \pi r$, with $N(\pi), N(r) > 1$. Thus $N(q) = N(\pi)N(r) \implies q^2 = N(\pi)N(r)$, and $N(\pi) = q$.
Let $\pi = a + b\omega$. Then,

$$\begin{aligned} N(\pi) &= a^2 - ab + b^2 = q \\ \implies 4q &= (2a-b)^2 + 3b^2. \end{aligned}$$

This implies that $q \equiv (2a-b)^2 \pmod 3$. If $3 \nmid q$, we have $q \equiv 1 \pmod 3$, for 1 is the only non-zero square modulo 3. It follows immediately that if $q \equiv 2 \pmod 3$, then q is a prime in $\mathbb{Z}[\omega]$. This completes the first case.

Case 2. In the second case, suppose that $p \equiv 1 \pmod 3$. Then, by Quadratic Reciprocity Law, we have,

$$\begin{aligned}\left(\frac{-3}{p}\right) = \left(\frac{-1}{p}\right)\left(\frac{3}{p}\right) &= (-1)^{\frac{p-1}{2}}\left(\frac{p}{3}\right)(-1)^{\frac{p-1}{2}\cdot\frac{3-1}{2}} \\ &= (-1)^{\frac{p-1}{2}}\left(\frac{p}{3}\right)(-1)^{\frac{p-1}{2}} \\ &= (-1)^{p-1}\left(\frac{p}{3}\right) \\ &= +1\cdot\left(\frac{p}{3}\right) \\ &= \left(\frac{1}{3}\right) \\ &= 1.\end{aligned}$$

Since $\left(\frac{-3}{p}\right) = 1$, -3 is a quadratic residue modulo p. Thus, $\exists a \in \mathbb{Z}$ such that

$$\begin{aligned}a^2 \equiv -3 \pmod p &\implies a^2 + 3 \equiv 0 \pmod p \\ &\implies a^2 + 3 = pb,\end{aligned}$$

for some $b \in \mathbb{Z}$. Thus, $pb = (a - \sqrt{-3})(a + \sqrt{-3})$. Thus, p divides $(a - \sqrt{-3})(a + \sqrt{-3}) = (a + 2\omega + 1)(a - 2\omega - 1)$.
If p were prime in $\mathbb{Z}[\omega]$, it would have to divide one of the factors, but this cannot happen since $p \neq 2$ and $2|p \notin \mathbb{Z}$. Thus, $p = \pi r$, where $N(\pi), N(r) > 1$

$$\implies p^2 = N(\pi)N(r).$$

Since π and r are not units, we consider the $N(\pi)$ case. We get,

$$\begin{aligned}p &= N(\pi) \\ \implies p &= \pi\overline{\pi}.\end{aligned}$$

Case 3. The third case is handled as follows.

$$\begin{aligned}x^3 - 1 &= (x-1)(x-\omega)(x-\omega^2) \\ \implies x^2 + x + 1 &= (x-\omega)(x-\omega^2).\end{aligned}$$

Put $x = 1$, then we get

$$\begin{aligned}3 &= (1-\omega)(1-\omega^2) \\ &= 1 - \omega^2 - \omega + \omega^3 \\ &= 1 - \omega^2 - \omega + 1 \\ &= 2 - \omega^2 - \omega \\ &= -(\omega^2 + \omega - 2),\end{aligned}$$

and

$$\begin{aligned}(1+\omega)(1-\omega)^2 &= (1+\omega)(1+\omega^2-2\omega)\\ &= 1+\omega^2-2\omega+\omega+\omega^3-2\omega^2\\ &= 2+\omega^2-\omega-2\omega^2\\ &= 2-\omega^2-\omega.\end{aligned}$$

Thus,

$$(1-\omega)(1-\omega^2) = (1+\omega)(1-\omega)^2 = -\omega^2(1-\omega)^2.$$

Upon taking the norm on both sides, we get,

$$9 = N(1-\omega)^2 \implies 3 = N(1-\omega).$$

Since $N(1-\omega) = 3$ and 3 is a prime in $\mathbb{Z}$, then by proposition 3.3, $1-\omega$ is a prime in $\mathbb{Z}[\omega]$. □

Just as in the ring $\mathbb{Z}$ and in the ring of all algebraic integers, the notion of congruence benefits $\mathbb{Z}[\omega]$. If $a, b, c \in \mathbb{Z}[\omega]$, and $c \neq 0$ is a non-unit, we say that $a \equiv b \pmod{c}$ if c divides $a-b$. Just as in $\mathbb{Z}$, the congruence classes modulo c can be made into a ring $\frac{\mathbb{Z}[\omega]}{c\mathbb{Z}[\omega]}$ called the *residue class ring modulo c*. The following result will be helpful in defining the residue class of $\mathbb{Z}[\omega]$.

3.2 Residue Class Ring of $\mathbb{Z}[\omega]$

Lemma 3.5. Let $\pi \in \mathbb{Z}[\omega]$ be a prime. Then $\frac{\mathbb{Z}[\omega]}{\pi\mathbb{Z}[\omega]}$ is a finite field with $N(\pi)$ elements.

Proof. First, we show that $\frac{\mathbb{Z}[\omega]}{\pi\mathbb{Z}[\omega]}$ is a field. Let $\alpha \in \mathbb{Z}[\omega]$ be such that $\alpha \equiv 0 \pmod{\pi}$. Since $\mathbb{Z}[\omega]$ is a Unique Factorisation Domain, there exist elements $\beta, \gamma \in \mathbb{Z}[\omega]$ such that $\beta\alpha + \gamma\pi = 1$. Thus, $\beta\alpha \equiv 1 \pmod{\pi}$, which shows that residue class of α is a unit in $\frac{\mathbb{Z}[\omega]}{\pi\mathbb{Z}[\omega]}$.
Next, to show that $\frac{\mathbb{Z}[\omega]}{\pi\mathbb{Z}[\omega]}$ has $N(\pi)$ elements, we consider the following cases:

Case 1. If q is a rational prime congruent to 2 modulo 3, then q is a prime in $\mathbb{Z}[\omega]$. Let $\pi = q$. We will prove that its complete set of representatives is $\{a + b\omega | 0 \leq a < q \, and \, 0 \leq b < q\}$. This will show that $\frac{\mathbb{Z}[\omega]}{q\mathbb{Z}[\omega]}$ has $N(\pi) = q^2$ elements. We begin the proof by assuming an element $\mu \in \mathbb{Z}[\omega]$.

Then since $\mathbb{Z}$ is an Euclidean Domain, $m = qs + a$ and $n = qt + b$, where $s, t, a, b \in \mathbb{Z}$ and $0 \leq a, b < q$.

$$\begin{aligned} \implies \mu &= qs + a + (qt + b)\omega \\ &= qs + a + qt\omega + b\omega \\ &= q(s + t\omega) + a + b\omega \\ &\equiv (a + b\omega) \pmod{q}. \end{aligned}$$

Thus, every element can be written as a member of the set $\{a + b\omega | 0 \leq a < q \, and \, 0 \leq b < q\}$.
Now, we have to show uniqueness. Suppose that $(a + b\omega) \equiv (a' + b'\omega) \pmod{q}$, where $0 \leq a, b, a', b' < q$. Then, $\frac{a-a'}{q} + \frac{b-b'}{q}\omega \in \mathbb{Z}[\omega]$. This implies that $\frac{a-a'}{q}, \frac{b-b'}{q} \in \mathbb{Z}$. This is only possible if $a = a'$ and $b = b'$.

Case 2. Now, we consider non-rational primes in $\mathbb{Z}[\omega]$. Suppose that $\pi \in \mathbb{Z}[\omega]$ where $N(\pi) = p$ and $p \equiv 1 \pmod{3}$. We prove that the complete set of representatives for $\frac{\mathbb{Z}[\omega]}{\pi\mathbb{Z}[\omega]}$ is given by the set $T = \{0, \ldots, p-1\}$. We need to prove that an arbitrary element $\alpha = m + n\omega \in \mathbb{Z}[\omega]$ is congruent modulo π to an element in T.
Suppose $\pi = a + b\omega$, where $a, b \in \mathbb{Z}$. We know that $\exists c \in \mathbb{Z}$ such that $cb \equiv n \pmod{p}$. Therefore, $(m+n\omega) \equiv m+cb\omega \pmod{p}$. Since $N(\pi) = \pi\overline{\pi} = p$, then, $(m+n\omega) \equiv m+cb\omega \pmod{\pi}$.
We rewrite α through a sequence of congruences $\pmod{\pi}$.

$$\begin{aligned} \alpha &= m + n\omega \\ &\equiv m + cb\omega \pmod{\pi} \\ &\equiv m + cb\omega - ca + ca \pmod{\pi} \\ &\equiv c(a + b\omega) + m - ca \pmod{\pi} \\ &\equiv c\pi + m - ca \pmod{\pi} \\ &\equiv m - ca \pmod{\pi}. \end{aligned}$$

This indicates that every element of $\mathbb{Z}[\omega]$ is congruent to a rational integer modulo π. But every rational integer is congruent to an integer in T modulo p and modulo π.
We have proved that every element of $\mathbb{Z}[\omega]$ is congruent to an element of $\{0, 1, 2 \ldots, p-1\}$ modulo π but we still need to prove that different elements in T represent distinct cosets.
If $t = t' \pmod{\pi}$ with $t, t' \in \mathbb{Z}$ and $0 \leq t, t' < p$. Then $\exists r \in \mathbb{Z}[\omega]$ such that

$$t - t' = \pi r \implies N(t - t') = N(\pi)N(r) \implies (t - t')^2 = pN(r) \implies p|(t - t')$$

and since $0 \leq t,\ t' < p$, we have $t = t'$.

Case 3. Now we consider the non-rational prime $\pi = 1-\omega$ in $\mathbb{Z}[\omega]$, with norm equal to three. We prove that the complete set of representatives for $\frac{\mathbb{Z}[\omega]}{\pi\mathbb{Z}[\omega]}$ is given by the set $S = \{0, 1, 2\}$. We need to prove that an arbitrary element $\alpha = a + b\omega$ is congruent modulo π to an element in S. Since $(1-\omega)|3$, we have the following for any $\alpha \in \mathbb{Z}[\omega]$.

1. $\alpha \equiv 0 \pmod{1-\omega}$ if and only if
   (a) $a \equiv 0 \equiv b \pmod 3$, or
   (b) $a \equiv 1 \pmod 3$, and $b \equiv 2 \pmod 3$.
2. $\alpha \equiv 1 \pmod{1-\omega}$ if and only if
   (a) $a \equiv 0 \pmod 3$, and $b \equiv 1 \pmod 3$, or
   (b) $a \equiv 1 \pmod 3$, and $b \equiv 0 \pmod 3$, or
   (c) $a \equiv 2 \pmod 3$, and $b \equiv 2 \pmod 3$.
3. $\alpha \equiv 2 \pmod{1-\omega}$ if and only if
   (a) $a \equiv 0 \pmod 3$, and $b \equiv 2 \pmod 3$, or
   (b) $a \equiv 2 \pmod 3$, and $b \equiv 0 \pmod 3$, or
   (c) $a \equiv 1 \pmod 3$, and $b \equiv 1 \pmod 3$.

Next, we need to prove that different elements in S represent different cosets. If $1 \equiv 0 \pmod{1-\omega}$, then we get $1 = u(1-\omega)$, where u is a unit. This implies that $N(u) = \frac{1}{3}$, which is a contradiction since u is a unit. We get the same result if we assume $2 \equiv 1 \pmod{1-\omega}$. If $2 \equiv 0 \pmod{1-\omega}$, then $2 = u(1-\omega)$. This implies that $N(u) = \frac{4}{3}$, which is a contradiction since u is a unit. Thus, different elements in S represent different cosets. □

Chapter 4

The Law of Cubic Reciprocity

Let π be a prime. As discussed in section 3.2, the multiplicative group of $\frac{\mathbb{Z}[\omega]}{\pi\mathbb{Z}[\omega]}$ has order $N(\pi)-1$. Hence, we have an analog of Fermat's Little Theorem.

Lemma 4.1. If $\pi \nmid \alpha$, then $\alpha^{N(\pi)-1} \equiv 1 \pmod{\pi}$.

Remark. If the norm of π is different from 3, then the residue classes of $1, \omega, \omega^2$ are distinct in $\frac{\mathbb{Z}[\omega]}{\pi\mathbb{Z}[\omega]}$. To see this, we will consider three cases.

Case 1. Assume 1 and ω are in the same residue class, i.e., $1 \equiv \omega \pmod{\pi}$. Then, $\pi | 1-\omega$, and since $1-\omega$ is a prime, we get that π and $1-\omega$ are associates. Thus,

$$N(\pi) = N(1-\omega)N(u) \implies N(\pi) = N(1-\omega) = 3.$$

This contradicts our assumption that the norm of π is different than 3.

Case 2. Assume ω and ω^2 lie in the same residue class. Then,

$$\omega \equiv \omega^2 \pmod{\pi} \implies \pi | \omega - \omega^2.$$

$\exists z \in \mathbb{Z}[\omega]$ such that

$$\begin{aligned} \pi z &= \omega - \omega^2 \\ N(\pi z) &= N(\omega - \omega^2) \\ N(\pi z) &= N(2\omega + 1) = 1 - 2 + 4 = 3. \end{aligned}$$

Thus, we get $N(\pi)N(z) = 3$. Since $N(\pi) \neq 3$, we get $N(\pi) = 1$. This is a contradiction as π is not a unit.

Case 3. Assume 1 and ω^2 are in the same residue class. Then,

$$1 \equiv \omega^2 \pmod{\pi} \implies \pi | 1 - \omega^2.$$

$\exists z \in \mathbb{Z}[\omega]$ such that

$$\begin{aligned} \pi z &= 1 - \omega^2 \\ \implies N(\pi)N(z) &= N(1-\omega)N(1+\omega) \\ \implies N(\pi)N(z) &= 3 \cdot 1 = 3. \end{aligned}$$

This implies that either $N(\pi) = 1$ or 3. Since π is not a unit, we get $N(\pi) = 3$. This contradicts our assumption that the norm of π is not 3.

Six complex numbers in $\mathbb{Z}[\omega]$ have a norm equal to 3. They are $1-\omega$, $-1-2\omega$, $-2-\omega$, $2+\omega$, $-1+\omega$, and $1+2\omega$. They are the associates of $1-\omega$. Since $\{1, \omega, \omega^2\}$ is a cyclic group of order 3, it follows that 3 divides the order of the multiplicative group $\left(\frac{\mathbb{Z}[\omega]}{\pi\mathbb{Z}[\omega]}\right)^*$, i.e., $3|N(\pi) - 1$, by Lagrange's Theorem.

Lemma 4.2. Suppose that π is a prime such that $N(\pi) \neq 3$ and that $\pi \nmid \alpha$. Then, $\exists$ a unique integer $m = 0, 1$, or 2 such that

$$\alpha^{\frac{N(\pi)-1}{3}} \equiv \omega^m \pmod{\pi}.$$

Proof. We know that π divides $\alpha^{N(\pi)-1} - 1$. Now,

$$\alpha^{N(\pi)-1} - 1 = \left(\alpha^{\frac{N(\pi)-1}{3}} - 1\right)\left(\alpha^{\frac{N(\pi)-1}{3}} - \omega\right)\left(\alpha^{\frac{N(\pi)-1}{3}} - \omega^2\right).$$

Since π is prime, it must divide one of the three factors on the right. According to the preceding remarks, it can divide at most one factor since if it divided two factors, it would divide the difference. This proves the proposition. □

4.1 Cubic Residue Character and the Law of Cubic Reciprocity

In this section, we define the Cubic Residue Character. The Cubic Residue Character plays the same role in the theory of cubic residues as the Legendre symbol plays in the theory of quadratic residues.

Definition 4.1 (Cubic Residue Character)**.** Let π be a prime in $\mathbb{Z}[\omega]$ with $N(\pi) \neq 3$. The ***cubic residue character*** of α modulo π is given by,

(a) $\left(\frac{\alpha}{\pi}\right)_3 = 0$ if $\pi|\alpha$,

(b) $\alpha^{\frac{N(\pi)-1}{3}} \equiv \left(\frac{\alpha}{\pi}\right)_3 \pmod{\pi}$, with $\left(\frac{\alpha}{\pi}\right)_3$ equal to 1, ω, or ω^2.

This character plays the same role in the theory of cubic residues as the Legendre symbol plays in the theory of quadratic residues.

Lemma 4.3. The following are the properties of the cubic residue character.

1. $\left(\frac{\alpha}{\pi}\right)_3 = 1$ if and only if $x^3 \equiv \alpha \pmod{\pi}$ is solvable, i.e., if and only if α is a cubic residue.
2. $\alpha^{\frac{N(\pi)-1}{3}} \equiv \left(\frac{\alpha}{\pi}\right)_3 \pmod{\pi}$.
3. $\left(\frac{\alpha\beta}{\pi}\right)_3 = \left(\frac{\alpha}{\pi}\right)_3 \cdot \left(\frac{\beta}{\pi}\right)_3$.
4. If $\alpha \equiv \beta \pmod{\pi}$, then $\left(\frac{\alpha}{\pi}\right)_3 = \left(\frac{\beta}{\pi}\right)_3$.

Proof. Let γ be a generator of the group $\left(\frac{\mathbb{Z}[\omega]}{\pi\mathbb{Z}[\omega]}\right)^*$. Then, the order of $\gamma \pmod{\pi}$ is $N(\pi)-1$. Suppose $\left(\frac{\alpha}{\pi}\right)_3 = 1$. Then we have $\alpha^{\frac{N(\pi)-1}{3}} \equiv 1 \pmod{\pi}$ by definition. Writing $\alpha = \gamma^t \pmod{\pi}$ for some $t \in \mathbb{Z}$, we have, $\gamma^{t\cdot\frac{N(\pi)-1}{3}} \equiv 1 \pmod{\pi}$. Thus, $N(\pi)-1$ divides $t \cdot \frac{N(\pi)-1}{3}$, i.e., $3|t$. Therefore, $\alpha \equiv \gamma^t \equiv \left(\gamma^{\frac{t}{3}}\right)^3 \pmod{\pi}$. Conversely, if $\exists$ a $\beta \in \mathbb{Z}[\omega]$ such that $\alpha \equiv \beta^3 \pmod{\pi}$, we have

$$\begin{aligned} \alpha^{\frac{N(\pi)-1}{3}} &\equiv \beta^{N(\pi)-1} \pmod{\pi} \\ &\equiv 1 \pmod{\pi}. \end{aligned}$$

Therefore, $\left(\frac{\alpha}{\pi}\right)_3 = 1$, by definition. This completes the proof of (1). The proof of (2) follows by definition.

Next we have,

$$\begin{aligned} \left(\frac{\alpha\beta}{\pi}\right)_3 &\equiv (\alpha\beta)^{\frac{N(\pi)-1}{3}} \pmod{\pi} \\ &\equiv \alpha^{\frac{N(\pi)-1}{3}}\beta^{\frac{N(\pi)-1}{3}} \pmod{\pi} \\ &\equiv \left(\frac{\alpha}{\pi}\right)_3 \cdot \left(\frac{\beta}{\pi}\right)_3 \pmod{\pi}. \end{aligned}$$

This completes the proof of (3).

If $\alpha \equiv \beta \pmod{\pi}$, then

$$\begin{aligned} \left(\frac{\alpha}{\pi}\right)_3 &\equiv \alpha^{\frac{N(\pi)-1}{3}} \pmod{\pi} \\ &\equiv \beta^{\frac{N(\pi)-1}{3}} \pmod{\pi} \\ &\equiv \left(\frac{\beta}{\pi}\right)_3 \pmod{\pi}, \end{aligned}$$

and so,

$$\left(\frac{\alpha}{\pi}\right)_3 = \left(\frac{\beta}{\pi}\right)_3.$$

This completes the proof of (4). □

From now on, we will use the notation $f_\pi(\alpha) = \left(\frac{\alpha}{\pi}\right)_3$ to denote cubic characters. In the following lemma, we study the behavior of cubic characters under complex conjugation.

Lemma 4.4. The following are the properties of the cubic character under complex conjugation:

1. $\overline{f_\pi(\alpha)} = f_\pi(\alpha)^2 = f_\pi(\alpha^2)$,
2. $\overline{f_\pi(\alpha)} = f_{\overline{\pi}}(\overline{\alpha})$.

Proof. $f_\pi(\alpha)$ is by definition 1, ω, or ω^2, and each of these numbers squared equals its conjugate.
From the definition of cubic residue character, we have

$$\alpha^{\frac{N(\pi)-1}{3}} \equiv f_\pi(\alpha) \pmod{\pi},$$

and we get

$$\overline{\alpha}^{\frac{N(\pi)-1}{3}} \equiv \overline{f_\pi(\alpha)} \pmod{\overline{)}\pi}.$$

Since $N(\pi) = N(\overline{\pi})$, we can write

$$\overline{\alpha}^{\frac{N(\pi)-1}{3}} = \overline{\alpha}^{\frac{N(\overline{\pi})-1}{3}}$$

which implies that

$$\overline{\alpha}^{\frac{N(\overline{\pi})-1}{3}} \equiv f_{\overline{\pi}}(\overline{\alpha}) \pmod{\pi}.$$

Thus,

$$f_{\overline{\pi}}(\overline{\alpha}) \equiv \overline{f_\pi(\alpha)} \pmod{\pi}.$$

This completes the proof of (2). □

Corollary. *$f_q(\overline{\alpha}) = f_q(\alpha^2)$ and $f_q(n) = 1$ if n is a rational integer prime to q.*

Proof. Since $\overline{q} = q$, we have $f_q(\overline{\alpha}) = f_{\overline{q}}(\overline{\alpha}) = \overline{f_q(\alpha)} = f_q(\alpha^2)$. This gives the first relation.
Since $\overline{n} = n$, we have $f_q(n) = \overline{f_q(n)} = f_q(n^2)$. Since $f_q(n) \neq 0$ it follows that $f_q(n) = 1$. □

The corollary states that n is a cubic residue modulo q. Thus, if $q_1 \neq q_2$ are two primes congruent to 2 modulo 3, then we have

$$f_{q_1}(q_2) = f_{q_2}(q_1).$$

This is a special case of the Law of Cubic Reciprocity. To formulate the general law, we must introduce the idea of a "primary" prime.

Definition 4.2 (Primary prime)**.** If π is a prime in $\mathbb{Z}[\omega]$, we say that π is ***primary*** if $\pi \equiv 2 \pmod 3$. If $\pi = q$ is rational, this is nothing new. If $\pi = a + b\omega$ is a complex prime, the definition is equivalent to $a \equiv 2 \pmod 3$, and $b \equiv 0 \pmod 3$.

We need the notion of "primary" to eliminate the ambiguity caused by the fact that every non-zero element of $\mathbb{Z}[\omega]$ has six associates.

Lemma 4.5. Suppose that $N(\pi) = p \equiv 1 \pmod 3$. Among the associates of π, exactly one is primary.

Proof. Write $\pi = a + b\omega$. The associates of π are π, $\omega\pi$, $\omega^2\pi$, $-\pi$, $-\omega\pi$, and $-\omega^2\pi$. In terms of a and b, these elements can be expressed as

1. $a + b\omega$
2. $-b + (a - b)\omega$
3. $(b - a) - a\omega$
4. $-a - b\omega$
5. $b + (b - a)\omega$
6. $(a - b) + a\omega$

Since $p = a^2 - ab + b^2 \equiv 1 \pmod 3$, not both a and b are divisible by 3. By looking at parts (1) and (2), we can assume that $3 \nmid a$. Considering parts (1) and (4), we can further assume that $a \equiv 2 \pmod 3$. Under this assumption,

$$\begin{aligned} & p = a^2 - ab + b^2 \\ \implies & 1 \equiv 4 - 2b + b^2 \pmod 3 \\ \implies & b(b-2) \equiv 0 \pmod 3. \end{aligned}$$

If $3|b$, then $a + b\omega$ is primary. If $b \equiv 2 \pmod 3$, then $b + (b - a)\omega$ is primary.

To show uniqueness, assume that $a + b\omega$ is primary. By considering the congruence class of the first term in part (2) to part (5), we see that none of these expressions is primary. Neither is the expression in part (6) since the coefficient of ω, a, is not divisible by 3. □

We can now state the Law of Cubic Reciprocity.

Theorem 4.1 (The Law of Cubic Reciprocity)**.** *Let π_1 and π_2 be primary, $N(\pi_1)$, $N(\pi_2) \neq 3$, and $N(\pi_1) \neq N(\pi_2)$. Then*

$$f_{\pi_1}(\pi_2) = f_{\pi_2}(\pi_1)$$

Before we begin the proof, we make a few observations.

Remark. 1. There are three cases to consider. Namely, both π_1 and π_2 are rational, π_1 is rational and π_2 is complex, and both π_1 and π_2 are complex. The first case is trivial, as seen in the Corollary of lemma 4.4.

2. The cubic character of the units can be dealt with as follows, Since $-1 = (-1)^3$, we have $f_\pi(-1) = 1$ for all primes π.
   If $N(\pi) \neq 3$, then it follows from lemma 4.3, part (2), that $f_\pi(\omega) = \omega^{\frac{N(\pi)-1}{3}}$. Thus, $f_\pi(\omega) = 1, \omega$, or ω^2 according to whether $N(\pi) = 1, 4$, or 7 modulo 9.

3. The prime $1-\omega$ is a special case. If $N(\pi) \neq 3$, we would like to evaluate $f_\pi(1-\omega)$. This is done by Eisenstein in [3].

Theorem 4.2 (Supplement to the Cubic Reciprocity Law)**.** *Suppose that $N(\pi) \neq 3$. If $\pi = q$ is rational, write $q = 3k - 1$. If $\pi = a + b\omega$ is a primary complex prime, write $a = 3k - 1$. Then*

$$f_\pi(1 - \omega) = \omega^{2k}.$$

Proof. Let $\pi = q$ is a rational prime, i.e., $\pi = q \equiv 2 \pmod 3$, where $q = 3k - 1$. To begin with, we notice that since

$$(1 - \omega)^2 = -3\omega,$$

we obtain

$$\begin{aligned} f_q(1 - \omega)^2 &= f_q(-3\omega) \\ &= f_q(-3) \cdot f_q(\omega) \\ &= f_q(\omega). \end{aligned}$$

From remark (2) of theorem 4.1, we obtain,

$$\begin{aligned} f_q(\omega) &= \omega^{\frac{N(q)-1}{3}} \\ &= \omega^{\frac{q^2-1}{3}}. \end{aligned}$$

Thus,

$$\begin{aligned} f_q(1-\omega)^2 &= \omega^{\frac{q^2-1}{3}} \\ &= \omega^{\frac{9k^2-6k}{3}} \\ &= \omega^{3k^2-2k} \\ &= \omega^{-2k} \\ &= \omega^k, \end{aligned}$$

since $\omega^3 = 1$. Upon squaring both sides, we obtain

$$f_q(1-\omega)^4 = \omega^{2k} \implies f_q(1-\omega) = \omega^{2k}.$$

□

4.2 Proof of the Law of Cubic Reciprocity

Let π be a complex prime such that $N(\pi) = p \equiv 1 \pmod 3$. Since $\frac{\mathbb{Z}[\omega]}{\pi\mathbb{Z}[\omega]}$ is a finite field of characteristic p, it contains a copy of $\frac{\mathbb{Z}}{p\mathbb{Z}}$. Both $\frac{\mathbb{Z}[\omega]}{\pi\mathbb{Z}[\omega]}$ and $\frac{\mathbb{Z}}{p\mathbb{Z}}$ have p elements. Thus, we may identify the two fields. More explicitly, the identification is given by sending the coset of n in $\frac{\mathbb{Z}}{p\mathbb{Z}}$ to the coset of n in $\frac{\mathbb{Z}[\omega]}{\pi\mathbb{Z}[\omega]}$. This identification allows us to consider f_π as a cubic character on $\frac{\mathbb{Z}}{p\mathbb{Z}}$. If f is a cubic character, we recall the following results from the corollary of lemma 2.5, and lemma 2.6.

1. $g(f)^3 = pJ(f,f)$,
2. If $J(f,f) = a + b\omega$, then $a \equiv -1 \pmod 3$, and $b \equiv 0 \pmod 3$.

Since $J(f,f)\overline{J(f,f)} = p$, if $f \neq \epsilon$ and $f \cdot f \neq \epsilon$, the second assertion says that $J(f,f)$ is a primary prime in $\mathbb{Z}[\omega]$ of norm p. The next two lemmas give important results for proving the Law of Cubic Reciprocity. Assume π is primary.

Lemma 4.6. $J(f_\pi, f_\pi) = \pi$.

Proof. Let $J(f_\pi, f_\pi) = \pi'$. Since $\pi \cdot \overline{\pi} = p = \pi' \cdot \overline{\pi}'$, we have $\pi | \pi'$ or $\pi | \overline{\pi}'$. We wish to eliminate the latter possibility. From the definition of Jacobi Sums, we have

$$\begin{aligned} J(f_\pi, f_\pi) &= \sum_x f_\pi(x) f_\pi(1-x) \\ &\equiv \sum_x \left(x^{\frac{N(\pi)-1}{3}}\right)(1-x)^{\frac{N(\pi)-1}{3}} \pmod \pi \\ &\equiv \sum_x x^{\frac{p-1}{3}}(1-x)^{\frac{p-1}{3}} \pmod \pi \\ &\equiv \sum_x x^{\frac{p-1}{3}}(1-x)^{\frac{p-1}{3}} \pmod p, \end{aligned}$$

where the sum is over $\frac{\mathbb{Z}}{p\mathbb{Z}}$. It follows that the summation is equal to

$$0^{\frac{p-1}{3}} \cdot (1-0)^{\frac{p-1}{3}} + 1^{\frac{p-1}{3}} \cdot (1-1)^{\frac{p-1}{3}} + 2^{\frac{p-1}{3}} \cdot (1-2)^{\frac{p-1}{3}} + \cdots + (p-1)^{\frac{p-1}{3}} \cdot (1-(p-1))^{\frac{p-1}{3}},$$

which is congruent to 0 modulo p. This shows that $J(f_\pi, f_\pi) \equiv 0 \pmod{\pi}$, i.e., $\pi | \pi'$, and therefore since both π and π' are primary and their norm is equal to p, $\pi = \pi'$. □

Corollary. $g(f_\pi)^3 = p\pi$.

Proof. Using the result that $g(f)^n = f(-1)pJ(f,f)J(f,f^2)\dots J(f,f^{n-2})$ from lemma 2.5, we get,

$$g(f)^3 = pJ(f,f).$$

Upon substituting the value of $J(f_\pi, f_\pi)$ from the previous lemma, we get,

$$g(f_\pi)^3 = pJ(f_\pi, f_\pi) = p\pi.$$

□

We can now prove the Law of Cubic Reciprocity.

Proof of the Law of Cubic Reciprocity. We consider two cases. In the first case, we have one complex and one rational primary prime. In the second case, we have two complex primary primes.

Case 1.

$$\pi_1 = q_1 \equiv 2 \pmod 3$$
$$\pi_2 = \pi,$$

with $N(\pi) = p$.
Consider the relation

$$g(f_\pi)^3 = p\pi.$$

Raise both sides to the $\frac{q^2-1}{3}$ power and take congruence modulo q. We get,

$$g(f_\pi)^{q^2-1} \equiv (p\pi)^{\frac{q^2-1}{3}} \pmod q$$
$$\implies g(f_\pi)^{q^2-1} \equiv f_q(p\pi) \pmod q.$$

Since $f_q(p) = 1$, this leads to

$$g(f_\pi)^{q^2-1} \equiv f_q(\pi) \pmod q$$
$$\implies g(f_\pi)^{q^2} \equiv f_q(\pi) g(f_\pi) \pmod q. \tag{4.1}$$

We now analyze the left-hand side:

$$\begin{aligned} g(f_\pi)^{q^2} &= \left(\sum_{t=0}^{p-1} f_\pi(t)\zeta^t\right)^{q^2} \\ &\equiv \sum_{t=0}^{p-1} f_\pi(t)^{q^2}\zeta^{q^2 t} \pmod{q}. \end{aligned}$$

Since $q^2 \equiv 1 \pmod 3$ and $f_\pi(t)$ is a cube root of 1, we have

$$\begin{aligned} g(f_\pi)^{q^2} &\equiv \sum_{t=0}^{p-1} f_\pi(t)^{q^2}\zeta^{q^2 t} \pmod{q} \\ &\equiv \sum_{t=0}^{p-1} f_\pi(t)\zeta^{q^2 t} \pmod{q} \\ &\equiv g_{q^2}(f_\pi) \pmod{q}. \end{aligned} \tag{4.2}$$

Using the result, $g_a(f) = f(a^{-1})g(f)$, if $a \neq 0$ and $f \neq \epsilon$, we get,

$$\begin{aligned} g_{q^2}(f_\pi) &= f_\pi(q^{-2})g(f_\pi) \\ &= f_\pi(q)g(f_\pi). \end{aligned} \tag{4.3}$$

Upon combining equation (4.2) and (4.3), we get,

$$g(f_\pi)^{q^2} \equiv f_\pi(q)g(f_\pi) \pmod{q}. \tag{4.4}$$

Upon combining equation (4.1) and (4.4), we get,

$$f_\pi(q)g(f_\pi) \equiv f_q(\pi)g(f_\pi) \pmod{q}.$$

Multiplying both sides of the congruence by $\overline{g(f_\pi)}$, and using the relation $g(f_\pi)\overline{g(f_\pi)} = |g(f_\pi)|^2 = (\sqrt{p})^2 = p$, we get,

$$\begin{aligned} f_\pi(q)p &\equiv f_q(\pi)p \pmod{q} \\ \implies f_\pi(q) &\equiv f_q(\pi) \pmod{q}, \end{aligned}$$

implying that

$$f_\pi(q) = f_q(\pi).$$

This completes the proof of Case 1.

Case 2. Consider the case of two complex primes π_1 and π_2, where

$$N(\pi_1) = p_1 \equiv 1 \pmod 3$$
$$N(\pi_2) = p_2 \equiv 1 \pmod 3.$$

Since π_1 and π_2 are primary, even $\overline{\pi_1}$ and $\overline{\pi_2}$ are primary, and $p_1 = \pi_1\overline{\pi_1}$ and $p_2 = \pi_2\overline{\pi_2}$. Starting from the relation $g(f_{\overline{\pi_1}})^3 = p_1\overline{\pi_1}$, raising to the $\frac{N(\pi_2)-1}{3} = \frac{p_2-1}{3}$ power, we get,

$$g(f_{\overline{\pi_1}})^{p_2-1} = (p_1\overline{\pi_1})^{\frac{p_2-1}{3}},$$

and then taking congruences modulo π_2, we obtain

$$\begin{aligned} g(f_{\overline{\pi_1}})^{p_2-1} &\equiv f_{\pi_2}(p_1\overline{\pi_1}) \pmod{\pi_2} \\ \implies g(f_{\overline{\pi_1}})^{p_2} &\equiv f_{\pi_2}(p_1\overline{\pi_1})g(f_{\overline{\pi_1}}) \pmod{\pi_2} \end{aligned} \tag{4.5}$$

As in the previous case, we analyze the left-hand side of the equation. Upon expanding, we get

$$\begin{aligned} g(f_{\overline{\pi_1}})^{p_2} &= \left(\sum_{t\in\mathbb{F}_{p_1}} f_{\overline{\pi_1}}(t)\zeta^t\right)^{p_2} \\ &\equiv \sum_{t\in\mathbb{F}_{p_1}} f_{\overline{\pi_1}}(t)^{p_2}\zeta^{tp_2} \pmod{\pi_2} \\ &\equiv \sum_{t\in\mathbb{F}_{p_1}} f_{\overline{\pi_1}}(t)\zeta^{tp_2} \pmod{\pi_2} \\ &\equiv g_{p_2}(f_{\overline{\pi_1}}) \pmod{\pi_2}. \end{aligned} \tag{4.6}$$

Using the result, $g_a(f) = f(a^{-1})g(f)$, if $a \neq 0$ and $f \neq \epsilon$, we get,

$$\begin{aligned} g_{p_2}(f_{\overline{\pi_1}}) &= f_{\overline{\pi_1}}({p_2}^{-1})g(f_{\overline{\pi_1}}) \\ &= f_{\overline{\pi_1}}(p_2)^2 g(f_{\overline{\pi_1}}). \end{aligned} \tag{4.7}$$

Upon combining equations (4.5), (4.6), and (4.7), we get

$$f_{\pi_2}(p_1\overline{\pi_1})g(f_{\overline{\pi_1}}) \equiv f_{\overline{\pi_1}}(p_2)^2 g(f_{\overline{\pi_1}}) \pmod{\pi_2}.$$

Upon multiplying both sides with $\overline{g(f_{\overline{\pi_1}})}$, we get

$$f_{\pi_2}(p_1\overline{\pi_1})p_1 \equiv f_{\overline{\pi_1}}(p_2)^2 p_1 \pmod{\pi_2},$$

and then upon dividing both sides by p_1, we obtain,

$$\begin{aligned} f_{\pi_2}(p_1\overline{\pi_1}) &\equiv f_{\overline{\pi_1}}(p_2)^2 \pmod{\pi_2} \\ \implies f_{\pi_2}(p_1\overline{\pi_1}) &= f_{\overline{\pi_1}}(p_2)^2. \end{aligned} \tag{4.8}$$

Similarly, if we start from the relation

$$g(f_{\pi_2})^3 = p_2\pi_2,$$

and raise both sides to the power $\frac{p_1-1}{3}$, and take congruence modulo π_1 on both sides, we get,

$$\begin{aligned} g(f_{\pi_2})^{p_1-1} &\equiv (p_2\pi_2)^{\frac{p_1-1}{3}} \pmod{\pi_1} \\ &\equiv f_{\pi_1}(p_2\pi_2) \pmod{\pi_1}. \end{aligned}$$

This implies that

$$g(f_{\pi_2})^{p_1} \equiv f_{\pi_1}(p_2\pi_2)g(f_{\pi_2}) \pmod{\pi_1}. \tag{4.9}$$

Upon expanding the left-hand side, we get

$$\begin{aligned} \left(\sum_{t\in\mathbb{F}_{p_2}} f_{\pi_2}(t)\zeta^t\right)^{p_1} &\equiv \sum_{t\in\mathbb{F}_{p_2}} f_{\pi_2}(t)^{p_1}\zeta^{tp_1} \\ &\equiv \sum_{t\in\mathbb{F}_{p_2}} f_{\pi_2}(t)\zeta^{tp_1} \\ &\equiv g_{p_1}(f_{\pi_2}) \pmod{\pi_1}. \end{aligned} \tag{4.10}$$

Using the result, $g_a(f) = f(a^{-1})g(f)$, if $a \neq 0$ and $f \neq \epsilon$, we get,

$$\begin{aligned} g_{p_1}(f_{\pi_2}) &= f_{\pi_2}(p_1^{-1})g(f_{\pi_2}) \\ &= f_{\pi_2}(p_1)^2 g(f_{\pi_2}). \end{aligned} \tag{4.11}$$

Upon combining (4.9), (4.10), and (4.11), we get

$$f_{\pi_1}(p_2\pi_2)g(f_{\pi_2}) \equiv f_{\pi_2}(p_1)^2 g(f_{\pi_2}) \pmod{\pi_1}.$$

Upon multiplying both sides with $\overline{g(f_{\pi_2})}$, we get

$$\begin{aligned} f_{\pi_1}(p_2\pi_2)p_2 &\equiv f_{\pi_2}(p_1)^2 p_2 \pmod{\pi_1} \\ \implies f_{\pi_1}(p_2\pi_2) &\equiv f_{\pi_2}(p_1)^2 \pmod{\pi_1}. \end{aligned} \tag{4.12}$$

Since $N(\pi_1) = N(\overline{\pi_1})$, and $p_2 = \overline{p_2}$, using Lemma 4.4 we can conclude

$$f_{\pi_1}(p_2) = f_{\pi_1}(\overline{p_2}) = f_{\overline{\pi_1}}(\overline{p_2}) = f_{\pi_1}(p_2^2) = f_{\overline{\pi_1}}(p_2)^2.$$

Thus,

$$f_{\pi_1}(p_2) = f_{\overline{\pi_1}}(p_2)^2. \tag{4.13}$$

From equation (4.8), we get

$$f_{\pi_2}(p_1\overline{\pi_1}) = f_{\overline{\pi_1}}(p_2)^2.$$

Upon multiplying both sides with $f_{\pi_1}(\pi_2)$, and substituting the value obtained in equation (4.12), we get,

$$\begin{aligned} f_{\pi_2}(p_1\overline{\pi_1})f_{\pi_1}(\pi_2) &= f_{\overline{\pi_1}}(p_2)^2 f_{\pi_1}(\pi_2) \\ &= f_{\pi_1}(p_2)f_{\pi_1}(\pi_2) \\ &= f_{\pi_1}(\pi_2 p_2) \\ &= f_{\pi_2}(p_1^2) \\ &= f_{\pi_2}(p_1\pi_1\overline{\pi_1}). \end{aligned}$$

Finally, we get

$$f_{\pi_1}(\pi_2) = f_{\pi_2}(\pi_1).$$

This concludes the second case and completes the proof of the Law of Cubic Reciprocity.

□

4.3 The Cubic Character of 2

The Law of Cubic Reciprocity can be used to develop the theory of cubic residues in the same manner as the Law of Quadratic Reciprocity led to the development of the theory of the Legendre symbol.

In this section, we will highlight a special case. Namely, we shall ask for all primes $\pi \in \mathbb{Z}[\omega]$ for which 2 is a cubic residue.

First, we notice that $x^3 \equiv 2 \pmod{\pi}$ is solvable if and only if $x^3 \equiv 2 \pmod{\pi'}$ is solvable for any associate of π. Thus, we assume that π is primary. If $\pi = q$ is a rational prime, then $f_q(2) = 1$, so 2 is a cubic residue for all such primes.

From now on, $\pi = a + b\omega$ is a primary complex prime.

Proposition 4.1. *$x^3 \equiv 2 \pmod{\pi}$ is solvable if and only if $\pi \equiv 1 \pmod{2}$, i.e., if and only if $a \equiv 1 \pmod{2}$ and $b \equiv 0 \pmod{2}$.*

Proof. By the Law of Cubic Reciprocity, we have

$$f_\pi(2) = f_2(\pi).$$

The norm of 2 is $2^2 = 4$. Thus

$$\pi = \pi^{\frac{4-1}{3}} \equiv f_2(\pi) \pmod 2.$$

It follows that $f_\pi(2) = 1$ if and only if $\pi \equiv 1 \pmod 2$. □

Proposition 4.2. *If $p \equiv 1 \pmod 3$, then $x^3 \equiv 2 \pmod p$ is solvable if and only if there are integers C and D such that $p = C^2 + 27D^2$.*

Proof. ($\Rightarrow$) : If $x^3 \equiv 2 \pmod p$ is solvable, so is $x^3 \equiv 2 \pmod \pi$ and thus $\pi \equiv 1 \pmod 2$ by Proposition 4.1. We have

$$4p = A^2 + 27B^2,$$

where $A = 2a - b$, $B = \frac{b}{3}$. Since b is even, so are B and A. Let $D = \frac{B}{2}$ and $C = \frac{A}{2}$. Then $p = C^2 + 27D^2$.

Suppose, conversely, that $p = C^2 + 27D^2$. Then $4p = (2C)^2 + 27(2D)^2$. By uniqueness $B = \pm 2D$; i.e., B is even, and thus so is b. It follows that $\pi = a + b\omega \equiv 1 \pmod 2$, and $x^3 \equiv 2 \pmod \pi$ is solvable. Since $\frac{\mathbb{Z}[\omega]}{\pi\mathbb{Z}[\omega]}$ has $p = N(\pi)$ elements there is an integer h such that $h^3 \equiv 2 \pmod \pi$. It is now easy to show that $h^3 \equiv 2 \pmod p$. If $\pi | h^3 - 2$, then $\overline{\pi} | h^3 - 2$ and $\pi\overline{\pi} = p | (h^3 - 2)^2$. Consequently, $p | h^3 - 2$, and we are done with the proof. □

Example 4.1. Let $p = 7$. Then $x^3 \equiv 2 \pmod 7$ is not solvable since there are no integers C and D such that $7 = C^2 + 27D^2$.

On the other hand, $p = 31 = 2^2 + 27 \cdot 1^2$. Thus, $x^3 \equiv 2 \pmod{31}$ is solvable. Indeed, $4^3 \equiv 2 \pmod{31}$.

Chapter 5

An Application of the Law of Cubic Reciprocity

There are several important applications of the Law of Cubic Reciprocity, one of the primary ones being determining if the congruence

$$x^3 \equiv a \pmod{p}$$

is solvable for a given prime p.

Another significant application of cubic residues is deriving the Generalised Lucas-Lehmer test, which helps us identify whether a given number of the form $A.3^n \pm 1$ is prime. Pedro Berrizbeitia and T. G. Berry, motivated by Rosen's derivation of the Lucas-Lehmer Primality test via quadratic reciprocity [2], show how the theory of the cubic residue symbol leads to a very simple and conceptual derivation of generalized Lucas-Lehmer tests for the numbers $A.3^n \pm 1$.

5.1 Mersenne Primes and the Lucas-Lehmer Test

The *Lucas-Lehmer test* is a primality test for *Mersenne Numbers.*

Definition 5.1 (Mersenne Numbers)**.** Numbers of the form

$$M_n = 2^n - 1$$

without the primality requirement are called *Mersenne Numbers.*

Definition 5.2 (Mersenne Primes)**.** *Mersenne primes* are primes of the form

$$M_p = 2^p - 1,$$

for some prime p.

The significance of studying Mersenne primes is their close connection to perfect numbers. In the 4th century BC, Euclid proved that if 2^p-1 is a prime, then $2^{p-1}(2^p-1)$ is a perfect number. In the 18th century, Leonhard Euler proved that, conversely, all even perfect numbers have this form. This is known as the *Euclid-Euler theorem*. It is unknown whether there are any odd perfect numbers.

Many of the largest known primes are Mersenne primes because Mersenne numbers are easy to check for primality.

Remark. If M_p is a prime, then the exponent p is also a prime, whereas the converse is not true. The smallest counterexample is

$$M_{11} = 2^{11} - 1 = 2047 = 23 \cdot 89.$$

The *Lucas-Lehmer* test is an efficient primality test that helps us test the primality of Mersenne Numbers.

Definition 5.3 (Lucas-Lehmer Primality Test)**.** It can be shown that for any odd prime $p > 2$, $M_p = 2^p - 1$ is prime if and only if M_p divides s_{p-2}, where

$$s_i = \begin{cases} 4, \text{ if } i = 0 \\ s_i^2 - 2, \text{ otherwise.} \end{cases}$$

The number $s_{p-2} \pmod{M_p}$ is called the *Lucas-Lehmer residue* of p.

Example 5.1. M_7 is a prime. We can prove this using the Lucas-Lehmer test.

$$\begin{aligned}
s_0 &= 4 \equiv 4 \pmod{127} \\
s_1 &= s_0^2 - 2 = 4^2 - 2 \equiv 14 \pmod{127} \\
s_2 &= s_1^2 = 14^2 - 2 = 194 \equiv 67 \pmod{127} \\
s_3 &= s_2^2 = 67^2 - 2 = 4487 \equiv 42 \pmod{127} \\
s_4 &= s_3^2 = 42^2 - 2 = 1762 \equiv 111 \pmod{127} \\
s_5 &= s_4^2 - 2 = 111^2 - 2 \equiv 0 \pmod{127}.
\end{aligned}$$

5.2 Generalized Lucas-Lehmer Test

The Lucas-Lehmer test for primality of the Mersenne number $2^p - 1$ has been generalised by H. Williams to give primality test for numbers of the form $M.k^a \pm 1$, for derivation of the Lucas-Lehmer test itself.

In this chapter, we will discuss the proof of the Generalised Lucas-Lehmer test given by Pedro Berrizbeitia and T.G. Berry in their paper titled *"Cubic Reciprocity and Generalised Lucas-Lehmer Tests for Primality of $A.3^n \pm 1$"* [**4**]. It is motivated by Rosen's derivation of the Lucas-Lehmer test [2] via quadratic reciprocity. They use the theory of cubic residues to achieve a very simple and conceptual derivation of the generalised Lucas-Lehmer tests for the numbers $A.3^n \pm 1$.

Before beginning the theorem and proof, we must fix some notations.

Let $\omega = e^{\frac{2\pi i}{3}} = \frac{-1+\sqrt{-3}}{2}$. We denote the norm of $\mathbb{Q}(\omega) \to \mathbb{Q}$ by N, and the trace by Tr. Let f denote the cubic character on $\mathbb{Z}[\omega]$. Let $N = M.3^a \pm 1$, where M is even and, M is not divisible by 3. If $N = M.3^a + 1$, assume $N \neq \left(\frac{M}{2} \pm 1\right)^2$. First, we find a small rational prime l such that N is not a cube modulo l. This can be done by direct search since l is small. Note that $l \equiv 1 \pmod 3$, since if $l \equiv 2 \pmod 3$, everything is a cube modulo l. We know that it is precisely rational primes congruent to 1 modulo 3 which split in $\mathbb{Z}[\omega]$, so l splits in $\mathbb{Z}[\omega]$, and we can write $l = \pi\overline{\pi}$ for some irreducible π in $\mathbb{Z}[\omega]$. Let $\delta = \frac{\overline{\pi}}{\pi}$.

Theorem 5.1 (Generalised Lucas-Lehmer Theorem)**.** *Let $\{S_k\}$ be the sequence defined by*

$$S_i = \begin{cases} Tr\left(\delta^M\right), \text{ if } i = 0 \\ S_k^3 - 3S_k, \text{ if } i{=}k{+}1. \end{cases}$$

Suppose $\frac{M}{2} < 4 \cdot 3^a - 1$. Then N is prime if and only if

$$S_{a-1} \equiv -1 \pmod N.$$

Remark. $S_k = Tr(\delta^{M.3^k})$.

Proof. We will split the proof into two cases.

Case 1. We first give the proof for $N = M \cdot 3^a - 1$.

($\Rightarrow$): Suppose N is prime. By choice, N is not a cube modulo l and $l = N(\pi)$. It follows that N is not a cube modulo π, hence, $f_\pi(N) = \omega$, or ω^2. Since in this

case, $N \equiv 2 \pmod 3$, it is a primary prime in $\mathbb{Z}[\omega]$, and upon applying the Cubic Reciprocity Law, we find

$$f_\pi(N) = f_N(\pi).$$

This means that

$$f_N(\pi) = \omega, \, or \, \omega^2$$

also. Then, by the definition of cubic residue character, we get

$$\begin{aligned} f_N(\pi) &\equiv \pi^{\frac{N^2-1}{3}} \pmod N, \\ \implies \pi^{\frac{N^2-1}{3}} &\equiv \omega^i \pmod N, \end{aligned}$$

where $i = 1$, or 2. We can rewrite this as

$$\begin{aligned} \left(\pi^{N-1}\right)^{\frac{N+1}{3}} &\equiv \omega^i \pmod N \\ \implies \left(\frac{\pi^N}{\pi}\right)^{\frac{N+1}{3}} &\equiv \omega^i \pmod N. \end{aligned}$$

But $\pi^N \equiv \overline{\pi} \pmod N$, as can be seen by observing that complex conjugation must coincide with the Frobenius $\phi(N)$ of the Galois group $Q(\omega)$ over Q. Thus, we obtain

$$\begin{aligned} \implies \left(\frac{\overline{\pi}}{\pi}\right)^{\frac{N+1}{3}} &\equiv \omega^i \pmod N \\ \implies \delta^{\frac{N+1}{3}} &\equiv \omega^i \pmod N. \end{aligned}$$

Upon taking trace on both sides, we get,

$$Tr\left(\delta^{\frac{N+1}{3}}\right) = Tr\left(\delta^{\frac{M.3^a-1+1}{3}}\right) = Tr\left(\delta^{\frac{M.3^a}{3}}\right) = Tr\left(\delta^{M.3^{a-1}}\right) \equiv -1 \pmod N.$$

From the remark at the beginning of the proof, we have

$$S_k = Tr\left(\delta^{M.3^k}\right).$$

Thus, N prime implies that

$$S_{a-1} \equiv -1 \pmod N.$$

The recurrence $S_k^3 = S_{k+1} + 3S_k$ follows easily from the definition of S_k, using $N(\delta) = 1$, hence the sequence $\{S_k\}$ satisfies the recurrence given in the theorem and we are done.

$(\Leftarrow)$: Assume now that $S_{a-1} \equiv -1 \pmod N$. We shall show that this implies that any

prime divisor of N is greater than $\sqrt{N}$ which implies N is prime. Suppose that q is a prime divisor of N. We let τ be the prime in $\mathbb{Z}[\omega]$ such that $\tau = q$ if $q \equiv 2 \pmod 3$ or else $\tau\overline{\tau}$ if $q \equiv 1 \pmod 3$. Now, $S_{a-1} \equiv -1 \pmod N$ implies that

$$\begin{aligned} & S_{a-1} \equiv -1 \pmod \tau \\ \Longrightarrow\ & Tr(\delta^{M\cdot 3^{a-1}}) \equiv -1 \pmod \tau \equiv Tr(\omega^i) \pmod \tau \; for\, i = 1, \, or\, 2. \end{aligned}$$

This implies that

$$\delta^{M.3^{a-1}} \equiv \omega^i \pmod \tau.$$

This implies that δ^M has order 3^a modulo τ in the group $\left(\frac{\mathbb{Z}[\omega]}{\tau\mathbb{Z}[\omega]}\right)^*$. Thus, 3^a divides $|\frac{\mathbb{Z}[\omega]}{\tau\mathbb{Z}[\omega]}|^* = N(\tau) - 1$ which is $q-1$ or q^2-1 according to $q \equiv 1$ or $q \equiv 2 \pmod 3$. In the first case we get 3^a divides $q-1$, in the second case we get 3^a divides $(q+1)(q-1)$. Thus, we conclude that every prime q that divides M is of the type either 3^a+1 or 3^a-1. It follows that every divisor of N is of the type $t_1 \cdot 3^a + 1$ or $t_2 \cdot 3^a - 1$ for some $t \geq 1$. Now, by following the argument as in Lemma 1 [5], it follows that N is a prime integer.

Case 2. In this case we will give the proof for $N = M \cdot 3^a + 1$. The proof will be slightly different, since now $N \equiv 1 \pmod 3$ and therefore splits in $\mathbb{Z}[\omega]$.

($\Rightarrow$): Assume N is prime, and chose $\theta \in \mathbb{Z}[\omega]$ to be a primary prime such that $N = \theta\overline{\theta}$. As before, N is not a cube modulo l, hence not a cube modulo π, so $f_\pi(N) = \omega^i$, where $i = 1$, or 2. Then,

$$\begin{aligned} \omega^i = f_\pi(\theta\overline{\theta}) &= f_\pi(\theta) f_\pi(\overline{\theta}) \\ &= f_\theta(\pi) f_{\overline{\theta}}(\pi) \\ &= f_\theta(\pi) f_\theta(\overline{\pi})^{-1} \\ &= f_\theta\left(\frac{\pi}{\overline{\pi}}\right). \end{aligned}$$

Thus,

$$f_\theta\left(\delta^{-1}\right) = \left(\delta^{-1}\right)^{\frac{N(\theta)-1}{3}} = \left(\delta^{-1}\right)^{\frac{N-1}{3}} \equiv \omega^i \pmod \theta.$$

Taking traces,

$$Tr\left(\left(\delta^{-1}\right)^{\frac{N-1}{3}}\right) = Tr\left(\left(\delta^{-1}\right)^{\frac{M\cdot 3^a+1-1}{3}}\right) = Tr\left(\left(\delta^{-1}\right)^{M\cdot 3^{a-1}}\right) \equiv -1 \pmod \theta.$$

Thus, we get,

$$S_{a-1} \equiv -1 \pmod \theta.$$

Interchanging θ and $\overline{\theta}$ in the calculation we have just made, we also obtain

$$S_{a-1} \equiv -1 \pmod{\overline{\theta}}.$$

Hence,

$$S_{a-1} \equiv -1 \pmod{N}$$

which is what we were supposed to prove.

($\Leftarrow$): Assume now that $S_{a-1} \equiv -1 \pmod{N}$. We shall show that this implies that any prime divisor of N is greater than $\sqrt{N}$ which implies N is prime. Suppose that q is a prime divisor of N. We let τ be the prime in $\mathbb{Z}[\omega]$ such that $\tau = q$ if $q \equiv 2 \pmod 3$ or else $\tau\overline{\tau}$ if $q \equiv 1 \pmod 3$. Now, $S_{a-1} \equiv -1 \pmod{N}$ implies that

$$\begin{aligned} & S_{a-1} \equiv -1 \pmod{\tau} \\ \Longrightarrow\ & Tr(\delta^{M \cdot 3^{a-1}}) \equiv -1 \pmod{\tau} \equiv Tr(\omega^i) \pmod{\tau}\ for\ i = 1, or 2. \end{aligned}$$

This implies that

$$\delta^{M.3^{a-1}} \equiv \omega^i \pmod{\tau}.$$

This implies that δ^M has order 3^a modulo τ in the group $\left(\frac{\mathbb{Z}[\omega]}{\tau\mathbb{Z}[\omega]}\right)^*$. Thus, 3^a divides $|\frac{\mathbb{Z}[\omega]}{\tau\mathbb{Z}[\omega]}|^* = N(\tau) - 1$ which is $q-1$ or q^2-1 according to $q \equiv 1$ or $q \equiv 2 \pmod 3$. In the first case we get 3^a divides $q-1$, in the second case we get 3^a divides $(q+1)(q-1)$. Thus, we conclude that every prime q that divides M is of the type either $3^a + 1$ or $3^a - 1$. It follows that every divisor of N is of the type $t_1 \cdot 3^a + 1$ or $t_2 \cdot 3^a - 1$ for some $t \geq 1$. Now, by following the argument as in Lemma 1 [5], under the hypothesis $N \neq \left(\frac{M}{2} \pm 1\right)^2$, it follows that N is a prime integer. □

Chapter 6

The Law of Biquadratic Reciprocity

In this section, we will introduce the ring of Gaussian integers, $\mathbb{Z}[i]$, and state some of its properties (for detailed proofs, one can refer to [1]). It is useful in studying the theory of Quartic residues and hence defining the Law of Biquadratic Reciprocity. Like Eisenstein integers is to the theory of cubic residues, Gaussian integers is to the theory of quartic residues.

If $\alpha \in \mathbb{Z}[i]$, then $(\alpha) = \alpha\mathbb{Z}[i]$ is the principal ideal generated by α. In this section, a prime will mean a positive prime of $\mathbb{Z}$. Recall that $\mathbb{Z}[i]$ is a Euclidean ring. Thus if π is irreducible and $\pi|\alpha\beta$ then either $\pi|\alpha$ or $\pi|\beta$. If $N(\alpha) = \alpha\overline{\alpha}$ is the norm of α, then $N(\alpha) = 1$ if and only if α is a unit. From this we can see that the units of $\mathbb{Z}[i]$ are ± 1, $\pm i$. If π is irreducible then there is a prime $p \in \mathbb{Z}$ such that $\pi|p$. If $\alpha \in \mathbb{Z}[i]$, and $N(\alpha)$ is prime then α is irreducible.

The following lemma classifies the primes in $\mathbb{Z}[i]$.

Lemma 6.1. The primes in $\mathbb{Z}[i]$ can be classified into three types.

1. $p = 2$ is not a Gaussian prime. In fact, $p = 2 = (1+i)(1-i)$, where $(1+i)$ and $(1-i)$ are prime (conjugate primes).
2. If p is prime, $p \equiv 1 \pmod 4$ then there is an irreducible π such that $p = \pi\overline{\pi}$.
3. If $q \equiv 3 \pmod 4$ is a prime in $\mathbb{Z}$, then q is irreducible considered as an element of $\mathbb{Z}[i]$.

Definition 6.1. A non-unit $\alpha \in \mathbb{Z}[i]$ is primary if $\alpha \equiv 1 \pmod{(1+i)^3}$.

Remark. A non-unit α is primary if and only if either $a \equiv 1 \pmod 4$, $b \equiv 0 \pmod 4$ or $a \equiv 3 \pmod 4$, $b \equiv 2 \pmod 4$. We note that any non-unit $\alpha \equiv 1 \pmod 4 \in \mathbb{Z}[i]$ is primary. Furthermore if α is primary then $(1+i) \nmid \alpha$. If q is a real prime, $q \equiv 3 \pmod 4$ then $-q$ is a primary irreducible. A primary element can be written as the product of primary irreducibles.

Similar to the residue class ring $\frac{\mathbb{Z}[\omega]}{\pi\mathbb{Z}[\omega]}$ for irreducible $\pi \in \mathbb{Z}[\omega]$, we have the residue class ring $\frac{\mathbb{Z}[i]}{\pi\mathbb{Z}[i]}$ for an irreducible $\pi \in \mathbb{Z}[i]$. The residue class ring $\frac{\mathbb{Z}[i]}{\pi\mathbb{Z}[i]}$ is a finite field with $N(\pi)$ elements.

Let π be a prime in $\mathbb{Z}[i]$. Then the multiplicative group of $\frac{\mathbb{Z}[i]}{\pi\mathbb{Z}[i]}$ has order $N(\pi) - 1$. Hence we have an analog of Fermat's Little theorem.

Lemma 6.2. If $\pi \nmid \alpha$ then $\alpha^{N(\pi)-1} \equiv 1 \pmod \pi$.

Lemma 6.3. If $\pi \nmid \alpha$, $(\pi) \neq (1+i)$ there exists a unique integer j, $0 \leq j \leq 3$ such that

$$\alpha^{\frac{(N(\pi)-1)}{4}} \equiv i^j \pmod \pi.$$

The definition of the biquadratic (or quartic) residue character follows.

Definition 6.2 (Biquadratic (or Quartic) residue character)**.** If π is an irreducible, $N(\pi) \neq 2$, then the biquadratic (or quartic) residue character of α, for $\pi \nmid \alpha$, is defined by $f_\pi(\alpha) = i^j$ where j is determined by Lemma 6.3. If $\pi | \alpha$ then $f_\pi(\alpha) = 0$.

Remark. If $\pi \nmid \alpha$ then $f_\pi(\alpha) = 1$ if and only if $x^4 \equiv \alpha \pmod \pi$ has a solution in $\mathbb{Z}[i]$.

Similar to the theory of cubic residues where we had that if $q \equiv 2 \pmod 3$ is a prime, then $f_q(a) = 1$, for $a \in \mathbb{Z}$, $q \nmid a$, we have that if q is a prime, $q \equiv 3 \pmod 4$, then, $f_q(a) = 1$ for $a \in \mathbb{Z}$, $q \nmid a$.

The quartic residue character is generalized as follows.

Definition 6.3. Let $\alpha \in \mathbb{Z}[i]$ be a non-unit such that $(1+i) \nmid \alpha$, and $\beta \in \mathbb{Z}[i]$. Write $\alpha = \prod_i \lambda_i$ where λ_i is irreducible. If $(\alpha, \beta) = 1$, define $f_\alpha(\beta)$ by

$$f_\alpha(\beta) = \prod_i f_{\lambda_i}(\beta).$$

We will state two more important lemmas before stating the Law of Biquadratic reciprocity.

Lemma 6.4. Let $\alpha \in \mathbb{Z}$, $\alpha \neq 0$, and $a \in \mathbb{Z}$ be an odd non-unit. If $(a, \alpha) = 1$, then

$$f_a(\alpha) = 1.$$

Lemma 6.5. If $n \neq 1$ is an integer $n \equiv 1 \pmod 4$, then $f_n(i) = (-1)^{\frac{(n-1)}{4}}$.

The general law of biquadratic reciprocity may be stated as follows. Let λ and π be relatively prime primary elements of $\mathbb{Z}[i]$. Then

Theorem 6.1 (The Law of Biquadratic Reciprocity)**.**

$$f_\pi(\lambda) = f_\lambda(\pi)(-1)^{((N(\lambda)-1)/4)((N(\pi)-1)/4)}.$$

If λ and π are primary, where $\lambda = c + di$ and $\pi = a + bi$, it is simple to see that $((N(\lambda)-1)/4)((N(\pi)-1)/4)$ and $((a-1)/2)((c-1)/2)$ have the same parity so one may write

$$f_\pi(\lambda) = f_\lambda(\pi)(-1)^{((a-1)/2)((c-1)/2)}.$$

In other words if either π or λ is congruent to 1 modulo 4 then π and λ have the same biquadratic character. If however both are congruent to $3 + 2i$, then λ and π have opposite character in the sense that

$$f_\pi(\lambda) = -f_\lambda(\pi).$$

The proof of the Law of Biqadratic Reciprocity uses the notion of biquadratic residues, generalised Gauss sums, and Jacobi sums. For a detailed proof, one may look in [1].

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